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Image generation unlearning in vector quantized models

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To Fusilla

Abstract

0.1 Italiano

I modelli generativi text-to-image basati su rappresentazioni latenti vettoriali quantizzate (VQ) hanno raggiunto prestazioni significative, ma presentano rischi legati alla generazione incontrollata di contenuti indesiderati o eticamente problematici. Questa tesi propone l'adattamento dell'algoritmo Latent Code Replacement (LCR), originariamente sviluppato per modelli generativi di movimento umano, al dominio della generazione di immagini. La metodologia LCR consente l'unlearning concreto e mirato di specifici contenuti, identificando e sostituendo codici latenti associati a immagini tossiche o protette da copyright, senza necessità di riaddestramento del modello. Sono esplorate diverse strategie per stimare l'associazione tra codici e contenuti indesiderati utilizzando generazione automatica e annotazione delle immagini. Gli esperimenti condotti su modelli generativi pre-addestrati dimostrano che la tecnica proposta permette una rimozione poco efficace dei contenuti problematici, preservando parzialmente la qualità complessiva delle immagini generate.

Parole chiave: unlearning, text-to-image, NSFW, VQGAN, generazione di immagini, latent code replacement.

0.2 English

Vector Quantized (VQ) latent-based text-to-image generative models have achieved notable performance but introduce risks associated with uncontrolled generation of undesirable or ethically problematic content. This thesis proposes adapting the Latent Code Replacement (LCR) algorithm, initially designed for human motion generation, to the image generation domain. LCR provides targeted concrete unlearning by identifying and substituting latent codes associated with toxic or copyright-protected content without retraining the model. Strategies for estimating code-toxicity associations through automatic image generation and labeling are explored. Experiments on pre-trained generative models demonstrate that the proposed method effectively removes problematic content while largely preserving overall image quality.

Keywords: unlearning, text-to-image, NSFW, VQGAN, image generation, latent code replacement.

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Chapter 1

Introduction

In recent years, text-to-image (T2I) generative models have achieved remarkable results in producing high-quality and semantically accurate visual content from natural language descriptions. Older model [17;24;31] have given way to a new state of the art for image fidelity [9;14]. Although these models offer significant creative and commercial opportunities, they also introduce substantial risks: among them, the possibility of generating unsafe, toxic, or ethically problematic content en masse.

The uncontrolled synthesis of violent, sexual, or otherwise harmful content like ficticious news, poses critical challenges to the safety and alignment of generative AI applications: although the presence of this content on the internet has always been present, the proliferation of automated pipelines with which inhumane volumes of AI-slop is generated and distributed on all the majour social media platforms has made it ever-easier to produce harmful content, often without much human intervention in the process that goes between creating a prompt and publishing a viral post [5].

Furthermore, as Large T2I models require massive datasets to be trained and be commercially viable in the current ever escalating AI arms race, these datasets have been too massive to be thoroughly human supervised. As many foundational datasets have done [30] [32], all autonomously scraped from the internet and only later sections of the dataset are sanitized with much human intervention. Based even on the strictest and most advanced laws of Internet privacy, those laid out by the European GDPR [28], the impracticality of collecting consent from all the natural persons which appear in the data scraped was recognized; consequently, unless an individual wishes to be erased from a dataset, the creators of the dataset has no legal obligation to notify the owner of the usage of said data, they are obliged to state formally the use that that data will have and the level of de-anonymization necessary to carry out the task. Worryingly, even if one's presence in a dataset is successfully erased, what is to be done about all the models trained on the dataset before the erasure? In their parameters are nested information which should have been rightfully deleted, which could enable generative models to reconstruct otherwise erased data. It would be in the interest of those handling the distribution of these models in need of rightful erasure to have an alternative unlearning technique which does not involve retraining the model from scratch on the newly updated dataset.

Similarly companies, independent artists, have found their copyrighted works

present in datasets and their works being replicated without their explicit consent. This has a real world impact on those who make a living out of their intellectual property, which is now, for the first time in history, a few tokens away from being posted and profited off of by millions by forgers, who perhaps know not the name of the artist they are making a forgery of.

Now more than ever there is a growing need for mechanisms that allow developers to selectively "forget"[29] certain concepts or behaviors embedded in model weights or latent representations, many approaches have implemented an approximate unlearning: the resulting model is seemingly equivalent to a model which "forgot" some sensitive data, but in actuality undesirable information are still nested deep into the model's parameters, with enough adversarial prompting, and an exploration of the obfuscated embedding space, these seemingly forgotten pieces of information as captured by some emerging benchmarks[11][10] can resurface. A much more robust type of unlearning prevents many types of attacks by erasing or otherwise modify the parameters in such a way that data is no longer available to the model. The technique proposed and that will be discussed belongs to the second type of unlearning called concrete unlearning.

1.1 Latent Code Replacement

Latent Code Replacement (LCR)[6] algorithm, is a training-free concrete unlearning method originally proposed for human motion generation in the image generation domain. LCR was designed for generative models using a discrete latent space (such as those based on VQGAN or VQVAE[27], an architecture which is at the heart of models which were at the forefront of multimodal understanding and image generation. In these architecture the bottleneck introduces a particular non-linearity: latent vectors are mapped to a learned codebook of vectors and then passed to the encoder.

We adapt the LCR methodology to work in the image domain by proposing a modified strategy to identify and suppress latent codes associated with toxic or copyright/privacy protected content, without requiring access to original training data or perform a resource intense and time consuming fine-tuning of the model.

Main Contributions

- Proposed an adaptation of the LCR method to the image generation domain using discrete latent-space models.
- Explored different methods to estimate code-toxicity associations using prompt generation and automatic image labeling.
- We evaluate the effectiveness of the proposed approach on pre-trained generative models and analyze its impact on safety and image quality.

Chapter 2

Background and Related Work

2.1 Image Generation

The task of image generation via machine learning has emerged as a central problem in computer vision and artificial intelligence, aiming to produce high-quality, realistic images from learned data distributions. This domain has witnessed rapid progress due to advances in deep generative models, which leverage large datasets and complex architectures to model the deep and rich nature of visual data.

Seminal contributions to this field include the introduction of Generative Adversarial Networks (GANs) Goodfellow et al.[31], which framed the problem as a minimax game between a generator and a discriminator, and the development of Variational Autoencoders (VAEs) by Kingma and Welling[19], which approached generation from a probabilistic inference perspective. These foundational models paved the way for a range of generative methods that improved image fidelity, training stability, and controllability.

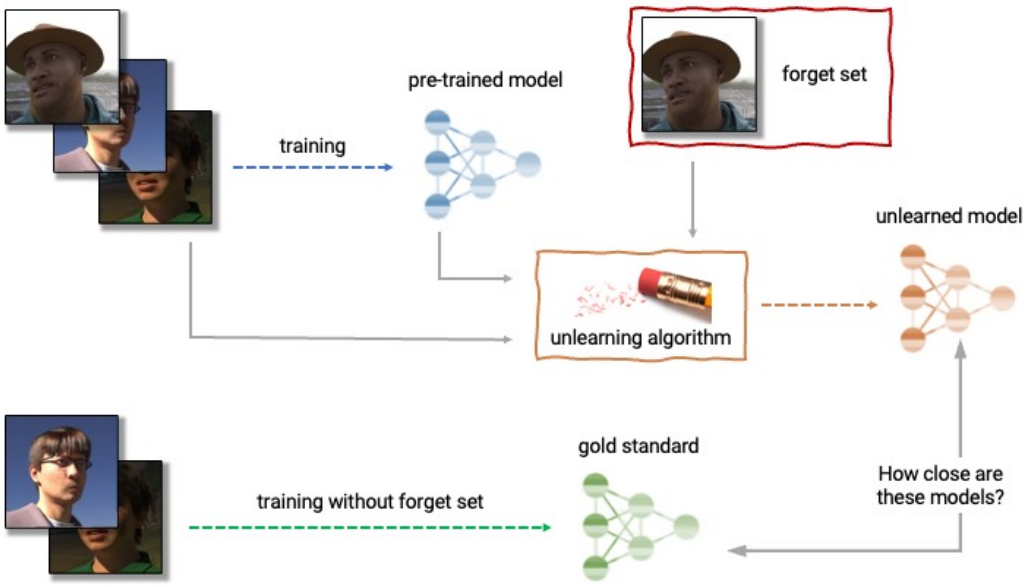
In recent years, diffusion models have redefined the state of the art in image generation. Techniques such as Denoising Diffusion Probabilistic Models (DDPMs)[25] and their improved versions, e.g., Imagen[22] and DALL · E 2[20], have demonstrated unprecedented performance in terms of photorealism, semantic coherence, and speed. These models operate by reversing a stochastic noising process, allowing for fine-grained control over the generation trajectory. Additionally, the integration of transformer-based architectures and large-scale text-conditioning, as in models like Stable Diffusion[21] and Midjourney, has expanded the creative and practical scope of generative models, enabling high-resolution image synthesis from natural language prompts.

2.2 Machine Unlearning in Generative Models

Machine unlearning refers to the selective removal of information from a trained machine learning model, often for privacy, ethical, or safety purposes. In the context of generative models, this involves removing specific concepts or features from the output space without compromising the model's general performance.

The idea behind many machine Unlearning algorithms is finding an efficient way of taking a pretrained model and making it indistinguishable to the model

Figure 2.1 The general idea behind machine unlearning



that was trained on the same dataset with the problematic data-points removed. This is often unfeasible, however many early attempts [26] relied on training an ensemble of models on non overlapping sections of a dataset, unlearning takes place when a number of models is not used during inference. Obviously this approach is not flexible, requiring a complete restructuring of the training of a model, but has been used with some success with LLMs [4]. Interestingly there has been a lot of work done on LLM's, a seminal work in this field was the Who's Harry Potter? [15] paper, which developed a very successful finetuning techniques based on three steps: First, a reinforced model is further trained on the target data to identify the tokens that are most related to the unlearning target, by comparing its logits with those of a baseline model. Second, idiosyncratic expressions are replaced in the target data with generic counterparts, leveraging the model's own predictions to generate alternative labels for every token. These labels aim to approximate the next-token predictions of a model that has not been trained on the target data. Third,the model was finetuned on these alternative labels, which effectively erase the original text from the model's memory whenever it is prompted with its context. The results appeared to demonstrate how targeted, and highly characterized concepts can be entirely removed from the model's parameters, with relatively quick finetuning.

2.2.1 Image unlearning

In the T2I domain, dominated by diffusion models, one of the most succesful approaches was Safe Latent Diffusion [18], a technique based on classifier-free guidance, which requires no finetuning and operates only at inference time: a safety threshold λ is selected and unconditioned vectors are nudged first by a prompt and later nudged in a safe direction. This safe direction was learned using an ad hoc collected dataset of inapporpriate image prompts (I2P). Although no real retraining was performed,

data had to be provided to inform the model what needed to be forgotten, and in a continuous embedded space this was represented in a direction, or rather a set of directions representing the shift between safe and unsafe concepts. In VQ architectures semantic meaning seems not to be as closely correlated to direction of embedded vectors as in their continuous counterpart [7], for our case the underlying assumption on how concepts are forgotten will have to change.

Other unlearning approaches required retraining on filtered datasets or parameter fine-tuning, both of which are computationally intensive. More recent techniques, such as Unified Concept Editing (UCE) [16] and Reliable and Efficient Concept Erasure (RECE). UCE enables fast and efficient unlearning for text-to-image models without requiring fine-tuning, modifying the cross-attention between text embeddings and latent image samples. Similarly RECE looks for the closest textual embedding $e\prime_f$ to an unlearned target concept e_f , then applies UCE to map e'_f to the safe target $\bar{e}_r$, this process is repeated for a pre-defined number of iterations, remapping neighboring toxic concepts to a safe prompt. The criterion of substituting toxic embeddings with safer alternatives is a great issue: as aforementioned in VQ architectures there is not a great semantic correspondence between codebook vector direction and its meaning, thus substituting up to now has been done with a random codebook vector, as more sophisticated techniques of substitution would require standalone research.

2.3 Discrete Latent Representations in Image Generation

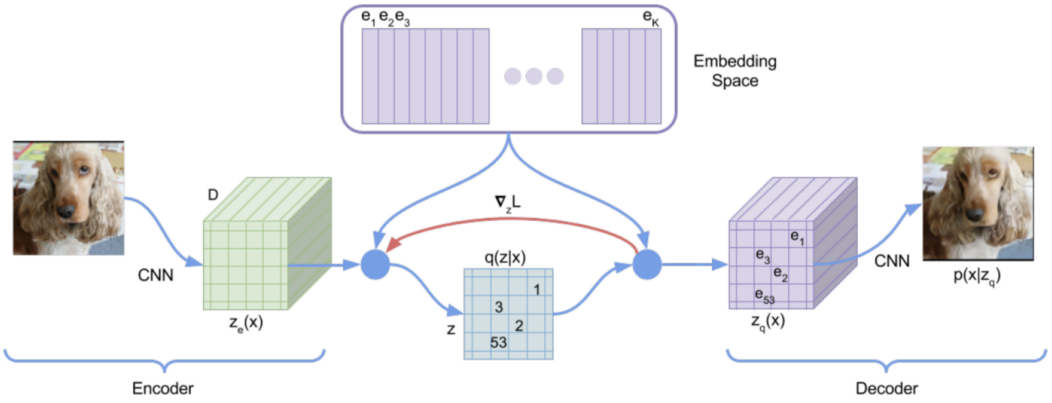
VQ-VAE reconstructs the input image by encoding it into a discrete latent space. Given an input Image $i \sim I$, an encoder $E(\cdot)$ maps it to a latent representation $Z \in \mathbb{R}^{\mathrm{d}}$ where d is the dimensionality of a latent embedding vector. Since Z is continuous, it is quantized to obtain a structured latent representation. This is done by mapping each latent vector z_t to the closest entry in a learned codebook $C = \{c_n\}_{n=1}^{N}$,where N is the number of discrete embeddings. The quantization step is defined as follows:

$$k = argmin||z - c_j||_2$$

The training objective of VQ-VAE consists of three components: i. the reconstruction loss ensures that decoded motion sequences remain close to the original input; ii. the commitment loss penalizes discrepancies between the encoder output and the selected codebook entries, encouraging stable quantization; iii. The codebook update loss adjusts the embeddings to better represent the encoded features, improving discretization over time.

Several modern generative models use discrete latent representations derived from Vector Quantized Variational Autoencoders (VQ-VAEs) or variants such as VQGAN and MOVQ. The two most famous of which are Deepseek's Janus [12] and OpenAi's DallE [20].

Figure 2.2 Architecture of a VQVAE, the base structure is that of a VAE except for the quantization non linearity in the bottleneck, since the operation of quantization is non differentiale during training gradient flow backpropagates from the decoder to the encoder bypassing the quantization operation entirely



2.3.1 DALLE

OpenAI's DALL · E2 [20] is a transformer-based generative model that synthesizes images from textual descriptions using a discrete latent space derived from VQ-VAE. Its core architecture extends the GPT framework to model sequences of text and image tokens jointly, treating image generation as an autoregressive prediction task over a unified vocabulary. Images are first compressed using a discrete VQ-VAE [27] or VQGAN encoder into a sequence of discrete tokens, which are then predicted autoregressively conditioned on textual input. During training, the model learns to map natural language prompts to plausible visual concepts by maximizing the likelihood of the corresponding tokenized image sequences.

A major strength of DALL · E2 lies in its ability to generate diverse and semantically rich images directly from text, this is made possible by large-scale transformer pretraining. The discrete latent space imposed by the VQ tokenizer allows compact, possibly more interpretable and expressive image representation, which enables natural image synthesis in a then-never-before-seen quality. Moreover, the use of a shared token sequence enables straightforward multimodal training without requiring separate decoders for text and images.

However, DALL · E2 also exhibits key limitations. The discrete nature of its latent space, while efficient, can limit fine-grained detail reconstruction compared to models operating in continuous spaces. Furthermore, its reliance on autoregressive decoding restricts parallelization during inference and can introduce coherence issues in larger images. Finally, because it was trained on large-scale, internet-scraped datasets, DALL · E2 can encode and reproduce biases present in the training data.

2.3.2 Janus

Janus uniquely decouples image encoding from its multimodal understanding and generation by employing a VQ tokenizer to embed images into the latent space of a unified autoregressive language model. This architectural decision allows Janus to

treat image and text tokens within the same representational framework, enabling seamless multimodal reasoning. The VQ tokenizer is initially pre-trained on ImageNet to capture a general-purpose visual vocabulary and later fine-tuned using curated text-image pairs sourced from prompt-based datasets[3], enhancing its alignment with natural language inputs for generative tasks.

During inference, both textual prompts and images are tokenized and embedded into a shared discrete latent space, which serves as input to a large language model. The model autoregressively generates image tokens conditioned on the input prompt, which are subsequently decoded via the fine-tuned VQ tokenizer into pixel-level outputs. This modular approach not only supports coherent image synthesis from complex prompts but also facilitates bidirectional understanding between modalities, enabling robust applications in retrieval, captioning, and visual reasoning.

While Janus no longer competes with state-of-the-art image generation models in terms of raw fidelity or photorealism, it remains a powerful tool for multimodal understanding due to its strong alignment between language and vision. Nevertheless, its training corpus—comprising largely unfiltered and scraped web data—introduces potential risks.

2.3.3 ConvVQGAN

Another interesting application of VQ models and transformers comes from a superresolution I2I ConvVQGAN[23]. A strong prior of convolutional networks is used to extract a rich codebook of informative visual parts, whose composition is arranged by an autoregressive transformer architecture. The model operates in two stages: first, a convolutional VQGAN learns to compress images into a discrete latent space, where each image is represented as a spatial grid of codebook entries selected from a learned dictionary. This stage leverages convolutional encoders and decoders, coupled with adversarial and perceptual losses, to ensure that the compressed representations retain high perceptual fidelity despite aggressive downsampling.

The second stage trains a transformer to model the composition of these latent tokens in an autoregressive fashion. This allows the model to capture long-range dependencies and global semantic coherence without being burdened by low-level details. By offloading the modeling of fine-grained visual structure to the convolutional VQGAN and focusing the transformer on arranging high-level concepts, the method achieves synthesis of high-resolution images with a never-seen-before quality (in the context of VQ models).

A critical advantage of this architecture lies in its scalability. Unlike pixel-space transformers, whose quadratic complexity severely limits their applicability to high-resolution data, ConvVQGAN reduces sequence length by encoding images at a lower spatial resolution with rich semantics. Furthermore, the convolutional stage introduces desirable inductive biases, such as spatial locality and translation invariance, which help guide the learning process and reduce the burden on the quadratic time complexity transformer. This hybrid formulation supports not only unconditional generation but also conditional synthesis tasks such as semantic image synthesis, pose-guided generation, segmentation and superresolution, all within a unified framework.

However, the approach is not without limitations. The quality of generated

samples is closely tied to the expressiveness of the codebook learned by the VQGAN; overly aggressive compression may lead to artifacts or loss of detail, which is extremely dependent on its hyperparameters (codebook count and downsampling factor).

Chapter 3

Methodology

3.1 Problem formalization

Unlearning undesirable concepts means preventing a model from generating images containing certain content, while preserving its capability to synthesize other. The problem is formalized as follows. Let us consider the dataset D consisting of paired texts T and images I, i.e. $D = \{(t, i)|t \in T, i \in I\}$. A T2I model parametrized by θ is a function f_θ able to synthesize novel images that fall in the same distribution as I from textual descriptions.

$$f_\theta(t \sim T) = i \sim I.$$

We define a forget set $D_f \subseteq D$ which contains only (t, i) pairs that we'd like to remove from the model; accordingly the retain set $D_r = D \setminus D_f$ of data pairs we consider safe. When unlearning D_f from the model f_θ , we mean reparameterizing $\theta \to \theta'$ such that the model will not be able to generate unsafe motions, while being able to still synthesize images from the retain dataset's distribution.

3.2 Latent Code Replacement (LCR)

Latent Code Replacement (LCR) is a training-free unlearning method introduced for human motion synthesis models. These models rely on a discrete latent space where motion sequences are represented as tokens quantized from a codebook. LCR identifies tokens highly correlated with toxic or harmful motions (e.g., kicking, punching) and replaces them with safe alternatives during inference, without altering model parameters destructively.

The proposed Latent Code Replacement unlearning technique, already tested on T2M models, is based on the key intuition that each code in the VQ-VAE's codebook encodes a specific motion, and that it is possible to identify toxic codes and replace them with safe ones. Unlike previous methodologies that modify attention parameters, LCR intervenes in the codebook's discrete latent space, enabling more precise control at the representational level under two basic assumptions:

1. codes represent disentangled concepts
2. undesirable concepts can be identified

Table 3.1 Some examples of midjourney prompt dataset copyright injection

Prompt undesired content injection	
Original prompt	modified prompt
A woman with short hair wearing glasses and **a man** is smiling. The background is	A woman with short hair wearing glasses and **shrek** is smiling. The background is ....
The first image shows an art gallery interior with a large illuminated **shrek** on the wall	The first image shows an art gallery interior with a large illuminated **sign** on the wall.

The former follows from the theory of VQ-VAE, where codebook discretization inherently encourages concept disentanglement [6]. This is further supported by empirical evidence: removing the codebooks associated with undesirable concepts does not degrade the generative performance, either quantitatively or qualitatively. Interestingly, the same codebooks can inject toxic behaviors, compromising otherwise safe images. This targeted intervention results in less disruption to the overall generation process while effectively removing undesirable patterns. By preserving the integrity of the learned parameters and selectively altering problematic codes in the latent space, this approach maintains generation quality for desirable images while successfully eliminating undesirable generations.

Intuitively there is no reason why this technique should not work in the image domain: the similarities in the architecture is present, the success in this otherwise completely different model might hint that LCR can be generalized to any architecture which embeds the input it recieves in a quantized latent space.

3.3 Dataset Construction

As previously discussed in order for our model to forget a concept it needs a dataset which explains what is to be removed. Two types of datasets were employed: undesirable content, full of negative examples and a sanitized dataset, containing data which align as closely to the distribution of data on which we would have wanted out model to be ideally trained (a dataset without undesirable content). It was thus as important to find a human curated and cleaned dataset as to find completely unsanitized ones.

Three types of dataset were prepared: prompts (I2P, SafeImagenetCaptions [8], Midjourney prompts), labeled images and image only (ImageNet [30], NSFW scraper [1])

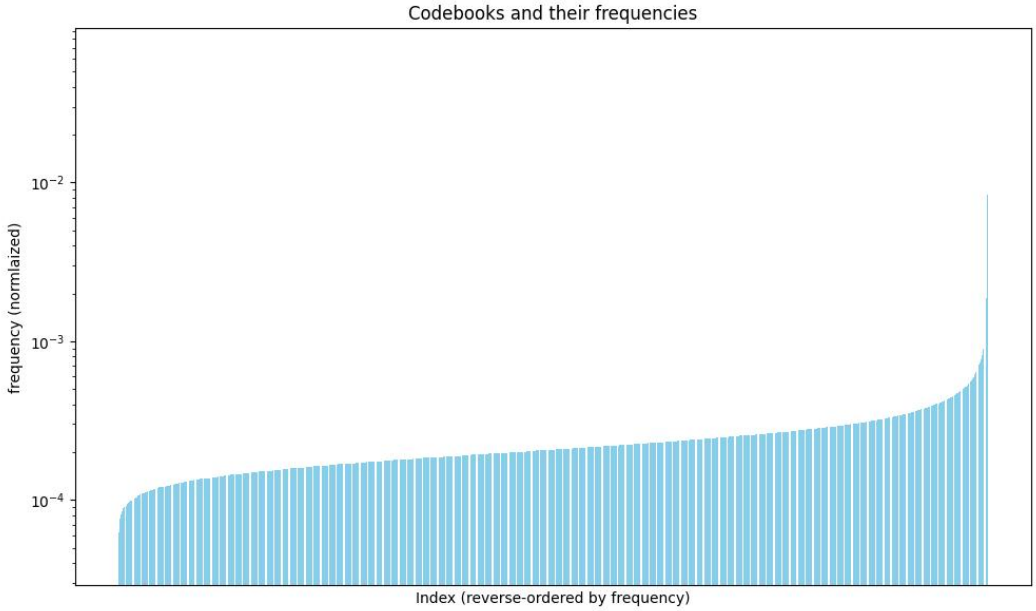
Several of these textual datasets were augmented with synthetic data: prompts with copyright infringement were created by injecting in place of the subject or the object of otherwise neutral prompts with some probability. This ensured that the generation would depict the same concept in however many differnet contexts possible, all to extract the most influencial notebooks which codified that information in the embedding space of the VQ model.

Inappropriate prompts were also refined and generated by other LLMs (often Janus itself), NSFW data labeled with Nudenet [2].

3.4 Selection of codebooks

Every model to be tested is run on these datasets, either images or prompts are provided to the model and the codebooks used for generation or reconstruction are annotated and cumulated on their frequency. This frequency is used to determine how involved a codebook is in the generation of a particular piece of content. Interestingly, based on empirical data it seems that on both generic data and highly class biased datasets use codebooks in an exponential fashion: some notebooks appear in almost every data-point reconstruction, while other hardly ever appear. To make a distinction between frequent codebooks and generally useful codebooks, quantized embeddings are scored with a metric called Forget to Retain Ratio, as proposed in the original LCR paper.

Figure 3.1 normalized frequencies of codebooks used for the reconstruction of a subset of Imagenet



What remains to be addressed are the exact datasets to be fed to the models: through training they have learned to generate undesirable content based on the biases and undesirable data in their training set. However some of these models did not have an openly available dataset, and thus providing the exact elements to be removed was not possible, as it might not be for models trained on multiple datasets: activations and values are going to be buried under the influence of several training rounds. The difference between providing never-seen-before data for codebook extraction and data-points present in the training set was investigated.

The idea behind using external datasets is that if a model is capable of reconstructing any image within, it will also be capable to generating similarly problematic content, the idea of LCR is that the model will not have access anymore to codebook entries used for undesired content generation, thus even the simple task of image reconstruction on undesirable content should be impossible. In this way never-seen-

before data from an undesirable dataset, is a way of providing negative examples to the VQ model. This does not conflict with the assumptions laid out for LCR.

3.4.1 Forget-to-Retain Ratio (FRR)

For each codebook entry c, its usage frequency is calculated in desirable and undesirable outputs as:

$$FRR(c) = \frac{N_f(c)}{N_r(c) + \epsilon}$$

Where $N_f(c)$ is the number of times code c appears in undesirable outputs and $N_r(c)$ in safe outputs. A smoothing constant ϵ is used to prevent division by zero. The codebooks thus found are ordered based on their FRR score and an hyper parameter k sets the number of latent vectors to be removed from the discrete codebook at the end of the lcr algorithm:

Algorithm 1 LCR

$K \leftarrow$
$\theta \leftarrow$
$C \leftarrow \theta.c$ ▷ codebooks
$N_f, N_r \leftarrow$
$L \leftarrow []$
for $c_i \in C$ **do**
  $L[i] \leftarrow [i, FRR(c)]$
end for
$L.sort(\lambda\ l \rightarrow -l[1])$ ▷ order bsaed on FRR score
$\Phi \leftarrow L[: K]$
for $c_i \in C$ **do**
  if $c_i \in \Phi$ **then**
    $\theta \leftarrow 0$
  end if
end for
$return\ \theta$

3.4.2 Entropy

Amongst the alternative metrics to FRR there is entropy: the idea is that some notebooks will appear always, no matter what image is generated, that codebook should have a low score as it is linked with general performances of the model, whereas a notebook that changes its frequency dramatically depending on the image that is being generated is a high entropy codebook and is probably linked to more specific concept. Given p the probability of a codebook being used in the decoding of a certain input:

$$p_i = \frac{num(c_i)}{C - 1}$$

where $num(c_i)$ is the number of times the codebook of index i was used to reconstruct a dataset which in total was reconstructed with C codes. The entropy score would be nothing different from the traditional definition of: $S = -p\log(p)$.

3.5 Code Replacement Strategy

Once toxic entries are identified, we apply LCR by substituting them with other alternatives. The simplest approach is to substutute with an arbitrary notebook (like we do in Algorithm 1), this would only serve to obfuscate with random noise and artifacts what is deemed undesirable. An alternative approach would be to substitute codebooks in the forget-set with more semantically meaningful codebooks in order to generate a "softer" unlearning. This is the difference between forgetting a copyrighted character's face and substituting it with a smear of color and covering up exposed nudity with reasonable obfuscation in the context of the rest of the image. For the purposes of this task both approaches are satisfactory, and in the experiments the first approach to replacement will be used by reason of simplicity.

Chapter 4

Experimental Setup

This chapter describes the models, datasets, evaluation criteria, and implementation details used to assess the effectiveness of the proposed LCR-based unlearning strategy in the T2I and I2I domain.

4.1 Hardware

All the experiments were carried out on 2 Nvidia 3090 GPUs, although useful to run experiments in parallel everything could have been done with a single GPU. This hardware was chosen as its architecture is compatible with required libraries and has enough VRAM to run an instance locally. It is important in these experiments that no safeguard is present and that code can be modified and run manually.

4.2 Models Used

The requirements for a model were the following:

- SOTA image generation, or close to
- VQ sub module and codebook learning
- Capable of producing either NSFW or copyrighted material
- Open-source weights and code

According to these specifications out of all the available models only these were suitable for experimenting with LCR:

- **Janus** — a VQ-based text-to-image model capable of generating NSFW content. Despite its relatively limited visual quality, it offers full access to codebook-based architecture and excellent multimodal inference.
- **VQGAN** — a super resolution transformer. It was chosen as in the category of VQ models it is by far the highest quality, its fast inference times support continuous testing

4.3 Data preparation

Experiments were run using 4 datasets:

1. **Midjourney prompt** — a collection of actual user's prompts to Midjourney, the classes of object represented are diverse both in style and in content, a subset 2000 prompts was randomly sampled and used

2. **Captioning of generated images** — to collect quantitative data with CLIP Janus was used to caption the images generated by itself. This was used to rewrite the text being used to generate the image into a format more fit to the use case of CLIP. 2000 captions were generated

3. **Shrek injected prompt** — starting from the dataset of Midjourney prompts the keyword "shrek" was injected strategically as explained in Section 3.3 to obtain varied prompts generating the undesired element. As the technique was very simple it was also quite error prone, thus more than 3000 prompts were created and used to generate images which were later scanned and cleaned of the ones lacking any presence of the, in this case, desired class with a simple image classifier to get around the 2000 image target

4. **Sanitized Imagenet** — a human cleaned version of Imagenet1K, all the images containing nudity, violence or other content which was not aligned with the autors code of conduct. It contains 1000 classes and 1.8 million rows. Various subsets were taken ranging from 1000 images to a 10000 images split.

5. **NSFW** — a 20.000 image dataset of sexually explicit images scraped off the internet, dataset was cleaned with NudeNet, to ensure that only negative examples were inputted to the model. Splits of 1000-10000 were generated from the dataset and used for the experiments



Figure 4.1 an example images generated flagged as nudity >0.6 (out of 1.0)

Other datasets were explored like I2P, but were used only for the final benchmark and not for LCR itself. The prompts for image generation proved not to be suitable to our T2I model, as Janus seemed to have problems creating NSFW imagery on its own, or rather, it could provide only low quality details of important human features, which made it difficult to establish a baseline upon which unlearning could be benchmarked. Moreover the dataset seemed to have been prepared to provide

a dataset to generic toxic content (nudity, self harm, hate, blood were some of the labels given to prompts), from 4.7k images only 854 were flagged as prompts with high nudity, of these 854 only 300 were actually used by Janus successfully, this few number of images was deemed too low for any kind of unlearning.

4.3.1 Hyper parameters for generation

Image using prompts was both conducted with high batch size and high temperature as a low batch size and temperature. The idea of high temperature and batch size is to utilize to the fullest a single prompt, by leaving the model free to generate the injected subject in the most varied poses, scenarios and interpretations, the experiments were run with the following parameters:

- **Temperature** = 1, **batches** = 6
- **Temperature** = 0.5, **batches** = 4
- **Temperature** = 0.1, **batches** = 1

The four majour Datasets were put against each other in pair, one dataset served as a neutral baseline and the other as a negative example, the pairs were Midjourney-Shrek T2I and Imagenet-NSFW for I2I. The couples were selected to serve as complementary pairs, mimicking forget and retain set: Midjourney prompt injected and none added a single element to the scrutiny of LCR, that being the injected subject, thus creating a robust example-counterexample data-pair. Same was the rationale behind the NSFW-ImageNET pair: the former provided negative examples and the latter not only served to keep intact all the codebooks not involved with explicit content generation, but also to maintain the original distribution of the ideal retain set of the model.

4.4 Codebook Tracing

For each image generated, codebook indices were extracted from the model's quantization layer and aggregated to construct the codebook frequency usage for a given dataset (an illustration of which is Figure. 3.1). These counts were ranked, in tandem with the codebooks of the other member of the corresponding pair, according to a selection metric (like FRR or entropy seen in Section 3.4), to compute a ranking of the most problematic codebooks. This approach allows for a simple iteration between codebook tracing strategies

4.5 Unlearning Procedure and Substitution policy

For ease of experimentation model weights were left unchanged and LCR was simulated at inference time by applying a filter to the codebooks extracted during the decoding before the construction of the final image. In reality the parameters of the model should be changed so that if an index is in the forget set then its value is changed to some predetermined value. It is not possible to alter this value depending

on the state of the decoded image but only as a fixed policy before inference. For example after having computed the most frequent quantized vectors (MFQV) a policy could be converting all the codebooks eliminated by LCR to the MFQV of the dataset, but not the MFQV in the image which inference is being made with. In our testing qualitative result changed significantly when forgotten vectors were set to 0 (or some infrequent, low activation vector) or set to the MFQV of the dataset (This is what we experimented with but there are alternatives as described in Section 3.5)

4.6 K values

The top-k most toxic codes (highest selection metric scores) were selected for replacement. Values of K were experimented with, and chosen to exemplify both quantitatively and qualitatively the effects of LCR. In the experiments several K values were discarded for a lack of any qualitative effect on the reconstructed images, the rage chosen for testing was raised until some unlearning started to manifest. Importantly, K values are dependent on the model's architecture and especially the number of codebooks each model will have: by studying the energy plot of the models it is clear how the majority of the information is held by the very top codebooks. For Janus the middle energy point is 4995 ± 46 out of a total of 16000 codebooks, while for VQGAN it is 752 ± 32 out of a total of 8192 (although for a single dataset like NSFW or ImageNet I have always observed less than 6000 active codebooks). Knowing this, a simple comparative benchmark was established: LCR would be compared with random removal of codebooks at the middle energy point, 5000 for janus and 800 for VQGAN. If the technique was att all successful it would have at least beat random removal.

4.7 Evaluation Metrics

The results of LCR were put against random code replacement, quantitative and qualitative scrutiny.

The practical results of the unlearning process will be self evident in the images generated after an application of the LCR algorithm to the pre-trained parameters of the two models used for our tests. The LCR codebook removal algorithm will be compared with a random removal of codes, this will serve as a baseline to verify the effective selection of toxic codebooks.

We used the following metric for quantitative evaluation:

- **CLIPScore** — measures semantic similarity between the text prompt and generated image (where applicable).

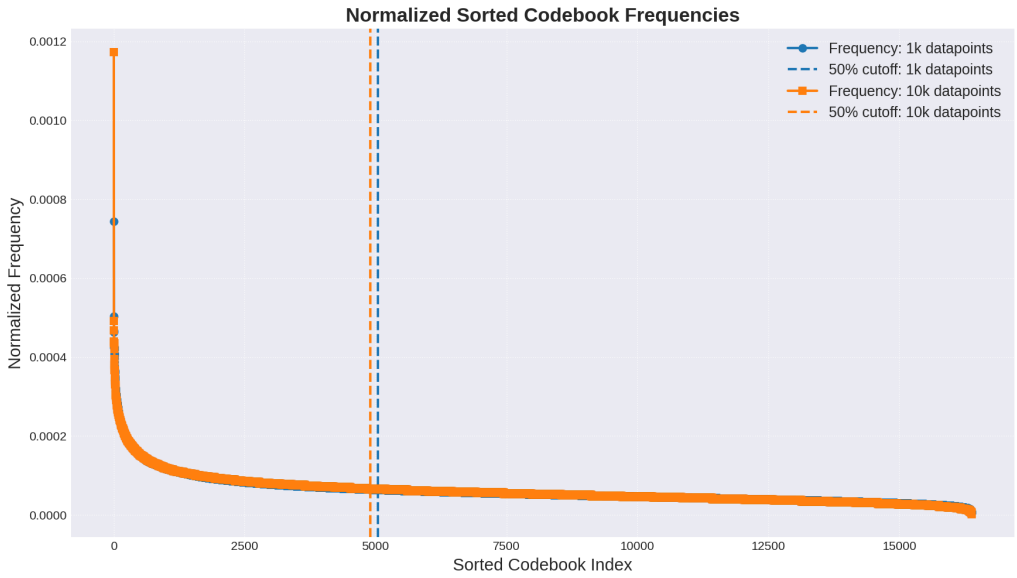


Figure 4.2 A comparison of the same dataset with increasingly many datapoints used for codebook tracing. Observe how the more data is used the less umbalanced the energy distribution becomes

Chapter 5

Results

5.1 Qualitative Evaluation

Qualitative analysis was conducted through visual inspection of image outputs before and after the application of LCR filtering. In the case of the **Shrek unlearning** experiment using the **Janus** model, LCR was found to be at first qualitatively successful. Post-unlearning generations revealed a consistent reduction in the presence of Shrek-like features across various injected prompts (Figure 5.2). Importantly, the images retained somewhat stylistic fidelity and coherence, first hinting that LCR effectively suppressed the target concept without degrading overall generative quality too aggressively. However, upon closer inspection and generation of more subtle and intentionally probing prompts, another conclusion was reached. Upon manual and classifier aided inspection of the 2000 generated images containing Shrek, only 10 still contained the discernible semblance of our forget target, however otherwise neutral images, not containing any trace of Shrek, were severely impacted (Figure 5.5).

5.1.1 Alternative substitutions

In the experiments run alternative substitution did not provide any benefits over the naive solution: substituting the MFQV in a dataset only served to obfuscate the forget target more spectacularly.



Figure 5.1 Qualitative results for images generated with prompts (injected) which generated images in the forget set; top K removal with LCR and FRR, MFQV substitution

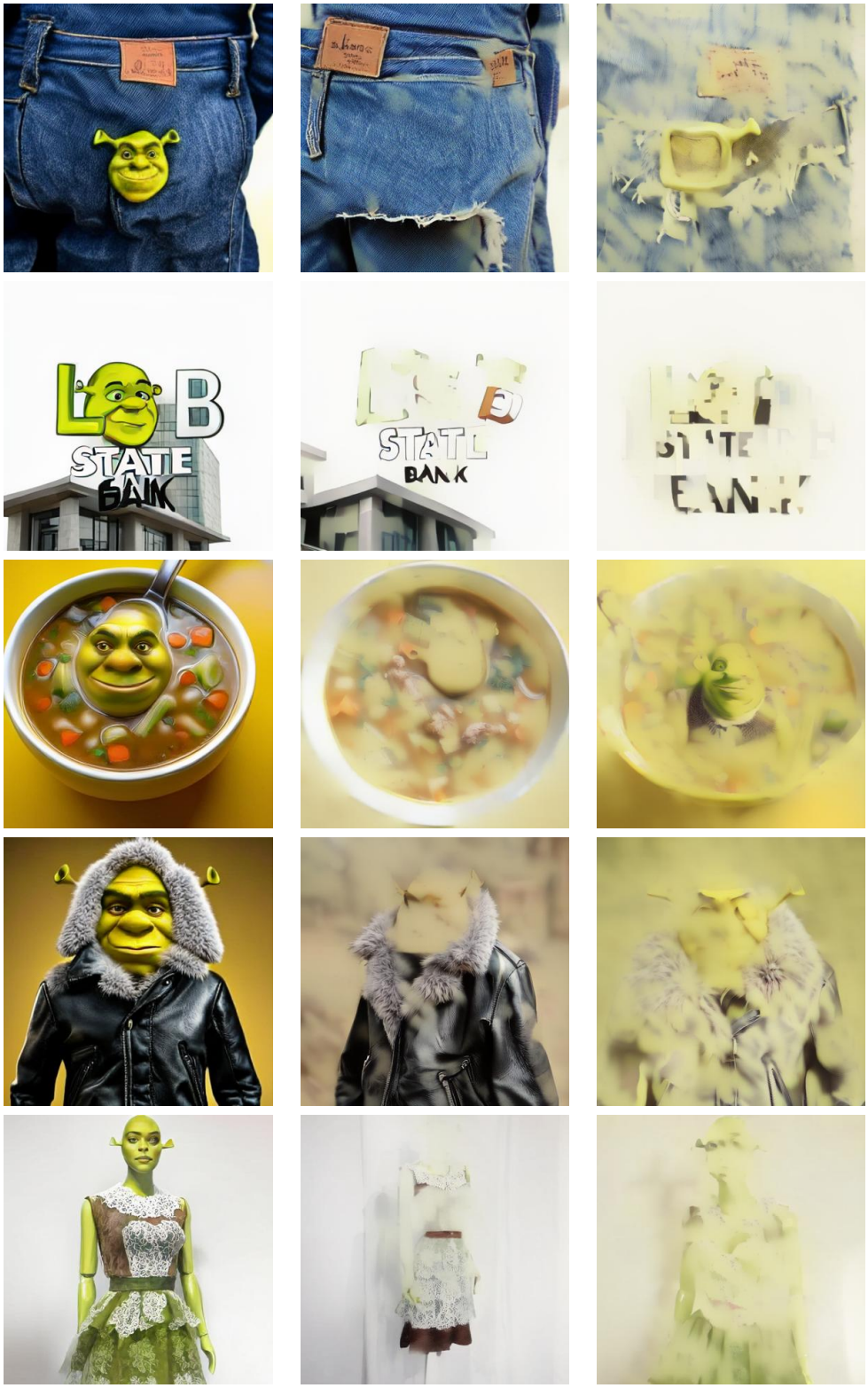


Figure 5.2 Qualitative results for images generated with prompts (injected) which generated images in the forget set and top K removal with LCR and FRR, random codebook removal

The same can be said for the NSFW-ImageNet pair, but with even less satisfying results: the model does correctly censor and distorts its forget objective, but also affects irreparably humanoids in general: parts of their skin is targeted by LCR which should not happen.



Figure 5.3 Qualitative results for the false positive removal in Janus I2I

In the case of **NSFW unlearning** on **VQGAN**, the results were poor. The collection and ranking of toxic notebooks was unsuccesful to distantiate itself from a removal of random codebooks, the dataset was increased to better infer the index of problematic notebooks but to no success.



LCR with K = 0 LCR with K = 200 random with K = 800

Figure 5.4 Qualitative results for the unsuccesful NSFW removal in VQGAN, random codebook removal on the right

Table 5.1 Evaluation of LCR on different model–dataset pairs. The ' symbol on the metric refers to LCR application. the ^{r},^{f} refer to the retain and forget datasets respectively. Error is 2*std

Model	**Dataset Pair**	$CLIP^{r}$ ↑	$CLIP^{r\,\prime}$ ↑	$CLIP^{f}$ ↑	$CLIP^{f\,\prime}$ ↑
Janus	Midjourney–Shrek	$34.8^{\pm 3.9}$	$29.8^{\pm 4.2}$	$26.2^{\pm 4.4}$	$19.5^{\pm 2.8}$
	Imagenet–NSFWscraper	$97.27^{\pm 2.8}$	$95.42^{\pm 2.9}$	$96.27^{\pm 1.8}$	$90.67^{\pm 2.9}$

5.2 Quantitative Evaluation

To validate these findings, the standard metric CLIPScore was used. These metrics offer complementary perspectives on the effectiveness of the unlearning: semantic alignment to prompts, and perceptual realism relative to real images. Although the impact of unlearning does not seem to have such a pronounced effect, LCR clearly targets D_f more than D_r. It is just not enough to define it as successful, which aligns perfectly with the qualitative analysis.

5.3 Discussion

- **Number of Codebooks**: According to the original LCR paper and the assumptions behind it, for a successful removal of an undesired concept only a few codebooks should be removed, lest LCR will jeopardize the output fidelity of the retain dataset as well. In this study no visible effect was achieved before removing $500 - 1000$ vectors from the Codebooks. This is a symptom that the technique for aggregating and ranking quantized vectors is not suited either for these models, or for this domain. The result of removing 2000 embedding vectors is that much more general features are captured and removed, as an example after having removed the top-2000 codebooks which should have been associated to shrek it becomes almost impossible to generate anything too green. Entropy as a policy was tried and failed to better in any way the performance of LCR.

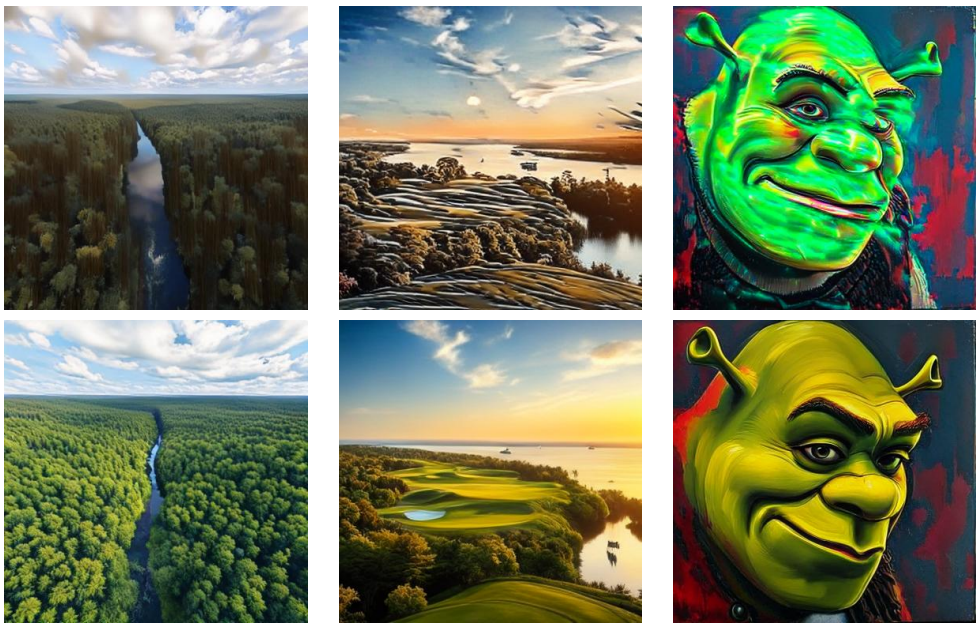


Figure 5.5 Qualitative results for the false negative and unsuccessful removals in Janus I2I, on the bottom pure reconstructions, on the top reconstructions with LCR K=2000.

- This experimental approach deviated a lot from the original paper: whereas

with T2M the whole dataset, used for the VQ training, was available, in these experiments all datasets were outsourced, but all perfectly capable of being generated by the VQ bottleneck either in I2I or T2I. Perhaps, more than a dataset similar in its distribution to the one used for VQ training, what really could have benefitted the codebook selection is a more robust retain set or a more specific forget set.

- In the exploration of the generated NSFW dataset with LCR a peculiar effect was found: as much of the unlearning translated from NSFW unlearning to skin unlearning, some of the individuals depicted in the dataset, who had darker shades of skin colour, were completely unaffected by the unlearning. Furthermore: male genitalia, which had given trouble to both models during LCR-less decoding-encoding, were the least targeted by the unlearning, despite having a skintone which more closely matched the apparent ethnic-bias in the NSFW dataset.

5.3.1 faster, better, stronger

As it was addressed in the discussion, the datasets could be made better:

- D_f **more specific**: the forget set used in this experiment was general. Images were generated with shrek in many contexts, NSFW images depicted a plethora of different individuals with different levels of skin exposure. To better the focus of the learned codebooks a dataset should have been composed of only the face of Shrek, or genitalia and nipples for NSFW. This way there would be less variability in the codebooks used by encoding images $\in D_f$. Less variability means higher frequency for those codebooks we are interested in forgetting.

- D_r **more robust**: In the Cleaned ImageNet counterpart to NSFW there were all different randomly sampled classes from the original dataset, which at first seemed like a strength, however: a more predominant presence of the class human could have made the embedding vectors "used" to the presence of skin in appropriate areas, thus getting rid of the false positive forgetting of faces and hands.

5.4 Future Work

There is certaintly many areas in which this technique could be expanded upon and bettered:

- addressing the issues brought up during result's discussion is certainly a priority. I would be confident that either making the retain set more robust or the forget set more specific would be enough to achieve a better unlearning

- Most of the models and datasets used were primarily very complex natural images and state of the art VQ models. Although other successful investigations of VQ concept/semantic learning have proven that VQ models are capable of disentangling concepts through quantization[13], how exactly this happens

is still not fully understood. A study on this same technique being applied in simpler contexts could provide a noise free environment in which more theoretical and in depth research can be performed on the semantic value of VQ codebooks.

- As it was highlighted in other works [7] [13] and other experiments carried out during this project, it is not yet clear how complex scenes like natural images are quantized and used for decoding in VQ models. The notion of proximity in space being linked to semantic meaning in the space of learned codebooks is not a meaningful metric, attempts to change codebooks based on cosine similarity instead of doing it randomly has proven no successful results. Improving the metric used for toxic codebook selection is of paramount importance in LCR. I would go so far as to modify assumption 2 of LCR stating that it is not sufficient that "undesirable concepts can be identified" but more precisely "undesirable concepts can be efficiently discriminated from desirable ones", like a perceptron dividing two datasclouds with a line the better a discriminatory metric we have the mor eeffective LCR will be.

- Once LCR will be deployed into the parameters of a publicly available model it will be needed to asses its forgetting capabilities through the employment of adversarial prompting techniques or handcrafted attacks, I am confident LCR will withstand such attacks by virtue of its deep application in model's parameters, however a thorough experimentation would surely provide useful insight on the effectiveness of LCR.

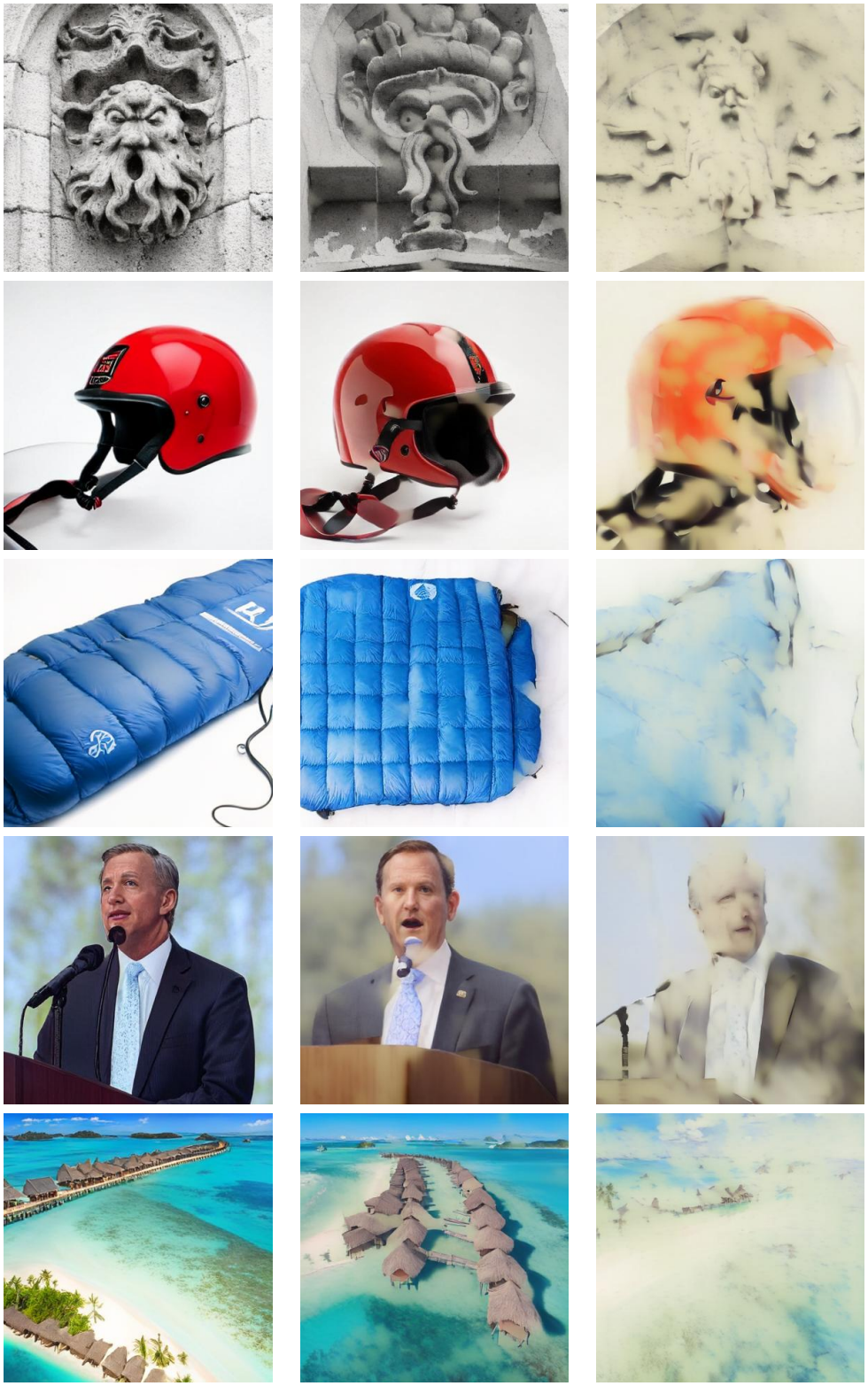


Figure 5.6 Qualitative results for images generated with prompts outside of the forget set (not injectd) and top K removal with LCR and FRR, random codebook removal

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